INFOSYS
3D MODELLING USING BLENDER

Internship Report submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN

COMPUTER SCIENCE AND ENGINEERING

By

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Department of Computer Science and Engineering
Accredited by NBA

Geethanjali College of Engineering and Technology
(UGC Autonomous)
(Affiliated to J.N.T.U.H, Approved by AICTE, New Delhi)
Cheeryal (V), Keesara (M), Medchal.Dist.-501 301.

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This is to certify that the Internship Report entitled **3D MODELLING USING BLENDER** is a Bonafide work done by **Akshad (23R11A0501)** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in "**Computer Science and Engineering**" from Jawaharlal Nehru Technological University, Hyderabad during the year 2025-2026

HOD - CSE

Dr. E Ravindra,

Professor

Examiner

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Designation:

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This certificate is proudly presented to

AKSHAD MISHRA

of Geethanjali College of Engineering and Technology (Roll No:23R11A0501) on successful completion of the Internship program at **Eeswar Solutions** from **05-05-2025** to **01-06-2025**. During this internship program he/she showed diligence, consistency and determination.

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DECLARATION BY THE CANDIDATE

I, Akshad Mishra, bearing Roll No. 23R11A0501, hereby declare that the Internship Report entitled **3D MODELLING USING BLENDER** is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**. This is a record of Bonafide work carried out by me in **INFOSYS** and the results embodied in this Internship report have not been reproduced or copied from any source. The results embodied in this Internship Report have not been submitted to any other University or Institute for the award of any other degree or Diploma.

Akshad Mishra

23R11A0501

CSE, GCET

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I would like to express my sincere gratitude to **Geethanjali College of Engineering and Technology**, Department of Computer Science and Engineering, for giving me the opportunity to undertake this internship project as part of the academic curriculum.

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I am deeply thankful to my faculty and mentors for their constant support, encouragement, and feedback throughout the project. I also appreciate the support of my classmates and friends who contributed to discussions and knowledge-sharing during this internship.

This experience has significantly enhanced my vivid thinking and problem-solving ability, and I am grateful for the opportunity to learn and grow through this project.

Akshad Mishra

23R11A0501

Introduction about Internship Organization

Infosys is known for its emphasis on innovation, quality, and sustainability. It has set up multiple Innovation Hubs globally and invests heavily in training, digital skills, and emerging technologies like AI, IoT, blockchain, and 3D modelling in sectors such as manufacturing, architecture, and education.

The internship I undertook was conducted through **Infosys in collaboration with Geethanjali College of Engineering and Technology**. It was designed to provide hands-on experience with practical designing of floor plans, objects and architectures using the **Blender application**

During this Internship students were acquainted with basic tools of blender such as gizmo, snaping, resizing, inserting objects, dimensions, angle, Edit and Object mode, meshes, rendering, lighting, extruding and so many other tools that helped us shape the objects their design.

This program proved to be a valuable learning experience by helping students understand the workflow of 3d modelling applications like Blender and AutoCAD which are majorly focused on architectural designs and animations.

Training Schedule

Weeks	Schedule
Week 1	• Introduction to 3D modelling • Applications for 3D modelling .

Week 2	• Learning Basic tools of Blender.
	  ○ Object mode
	  ○ Edit mode

Week 3	• Making real life objects
	    ○ Headphones
	    ○ Pencil
	    ○ Sandals
	    ○ Mug
	    ○ Coin
	    ○ Low poly house
	    ○ Water bottle
	    ○ Spoon
	    ○ Hammer

Week 4	• Final project
	    • Architecture
	        ○ Brutalist House Model
	.

ABSTRACT

3D modelling plays a pivotal role across industries such as architecture, product design, gaming, and animation—enabling realistic visualization and creative experimentation. This internship at Infosys focused on developing practical 3D modelling skills using **Blender**, an open-source, industry-standard software. Over four weeks, the program offered a structured immersion into core modelling techniques, beginning with the fundamentals of 3D space and progressing to advanced design applications. Early stages involved mastering Blender's interface and essential modes (Object and Edit Mode), followed by hands-on creation of real-world objects including headphones, mugs, tools, and furniture.

The capstone project featured the design of a **Brutalist-style architectural model**, incorporating geometric precision, scale realism, and aesthetic balance. Key technical skills acquired include mesh editing, object manipulation, topology management, and low-poly optimization—critical for efficient and visually compelling designs. This internship not only enhanced creative and technical proficiency but also offered insights into industry workflows and the practical applications of 3D modelling in professional environments. Future scope includes deeper exploration of texturing, lighting, animation, and integration with simulation or virtual reality platforms, further expanding the potential of Blender in modern digital design ecosystems.

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INTRODUCTION

As part of my internship, I explored 3D modelling and design using Blender, a powerful open-source graphics software. The internship focused on developing fundamental and intermediate skills in digital modelling through practical projects, starting from basic objects and culminating in a fully furnished architectural layout. Blender's versatile toolset and intuitive interface allowed me to learn industry-relevant techniques in mesh modelling, viewport shading, lighting, and scene composition.

During the initial weeks, I worked on primitive object manipulation and learned how to use modifiers, apply transformations, and model tools like hammers and furniture items. Over time, I expanded my knowledge by creating more complex 3D scenes such as house layouts, garages, and interior setups. To enhance realism and save time, I also utilized freely available asset libraries and Blender add-ons—such as the IKEA Furniture Library and **BlenderKit**—which provided licensed models for furniture, cars, and decor.

My final project involved designing a complete 3D house interior with realistic elements including furniture, cars, and textured floors. It demonstrated my ability to structure large scenes, organize objects across collections, and manage materials, lighting, and rendering settings. Screenshots from different stages of modeling and viewport modes (wireframe, solid, and material preview) reflect the evolution of my work.

This internship has significantly strengthened my understanding of 3D modelling and its practical applications in architectural visualization. It provided a solid foundation for pursuing more advanced workflows in visual design, game asset creation, or digital animation.

2. Related Work / Technologies Used

2.1 Related Work

3D modelling has seen rapid advancements in recent years, playing a critical role in fields ranging from architecture and industrial design to gaming and digital media. The use of open-source tools like **Blender** has democratized access to professional-grade modelling software, allowing creators and developers to design, prototype, and visualize complex structures with minimal cost overhead. In architecture, 3D modelling is frequently employed to present conceptual designs, simulate environments, and refine spatial aesthetics before physical execution. Similarly, industries such as product design and animation leverage 3D assets for digital prototyping and storytelling.

Previous studies and industrial practices highlight the importance of low-poly modelling for real-time applications and emphasize the role of accurate topology and mesh optimization in ensuring performance and visual fidelity. While sculpting and animation add further layers of realism and movement, solid proficiency in object modelling, texturing, and rendering form the backbone of most professional workflows. This internship builds on those foundations, offering a focused exploration into hard-surface modelling, scene composition, and architecture visualization using Blender.

2.2 Technologies Used

The internship primarily revolved around the **Blender 3D suite**, supplemented by a structured approach to learning and applying modern digital design workflows. Below are the key tools and concepts covered:

2.2.1. Blender

Blender served as the core platform for all modelling tasks. Its robust suite of features—including mesh editing, modifiers, viewport shading, and material nodes—enabled the creation of high-quality 3D assets. The use of Object Mode and Edit Mode allowed for geometric manipulation and refinement of forms.

2.2.2. Modelling Techniques

A variety of real-life objects were modelled using hard-surface techniques, focusing on geometry accuracy, proportional design, and clean topology. Key tools included:

- **Loop cuts and knife tools**
- **Extrude, inset, bevel, and mirror modifiers**
- **Snapping and alignment tools**

2.2.3. Texturing and Materials

Basic **texturing and material setup** was explored using Blender's Shader Editor, allowing realistic surface properties to be applied. Concepts like:

- **Base colour, roughness, metallic maps**
- **UV unwrapping**
- **Material assignment to individual faces**
  were incorporated into final designs.

2.2.4. Scene Composition and Rendering

Interns learned to arrange objects in a 3D scene, apply lighting (point, area, and sun lamps), and render final images using **Cycles** or **Eevee** render engines. Emphasis was placed on camera framing, object hierarchy, and background environment settings.

2.2.5. Final Project Tools

The **Brutalist architectural model** developed in the final week required application of modular modelling, array modifiers, and Boolean operations. Precision in grid snapping, real-world scaling, and architectural reference matching were emphasized.

2.2.6. Development Environment

All projects were created and managed within Blender's interface, with regular versioning and screenshots maintained for documentation. Interns also practiced exporting models in formats like **.FBX /.OBJ** or **.Blend** for compatibility with other pipelines or platforms.

2.2.7. Blender Add-ons and Asset Libraries

To enhance productivity and speed up the modelling process, several **free Blender add-ons and asset libraries** were utilized during the internship. These tools provided pre-built models, modifiers, and templates for furniture, structural elements, and household objects—allowing for quicker scene composition and more time spent on refinement and texturing. Some of the commonly used or accessible libraries include:

- **BlenderKit**
  An integrated asset library offering a large collection of **free 3D models, materials, HDRIs, and brushes**. Widely used for quick furniture additions and environment dressing. All assets are searchable directly within Blender.
- **Sketchfab Importer**
  Enabled access to thousands of **Creative Commons 3D assets** from Sketchfab. Useful for importing props or placeholder elements into the scene.
- **Archipack (Lite)**
  Provided basic **architectural primitives** such as walls, windows, stairs, and floors—helpful during the development of the Brutalist house model.

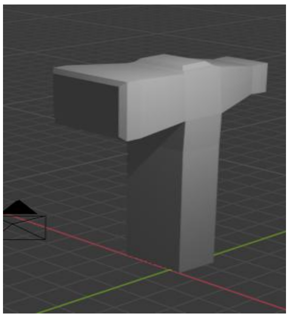
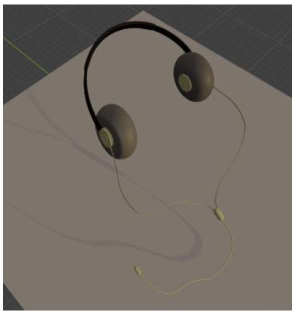
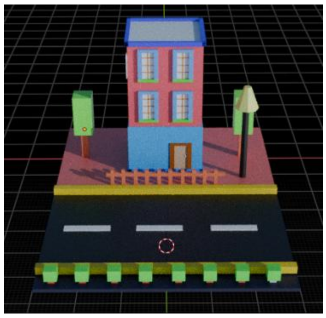
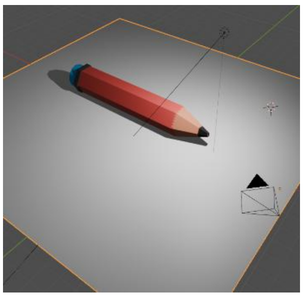
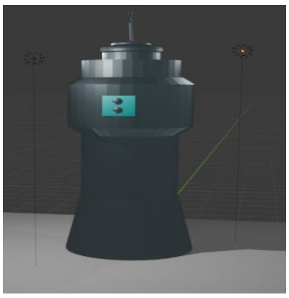
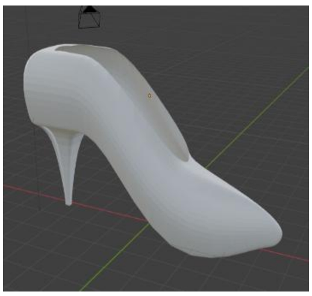
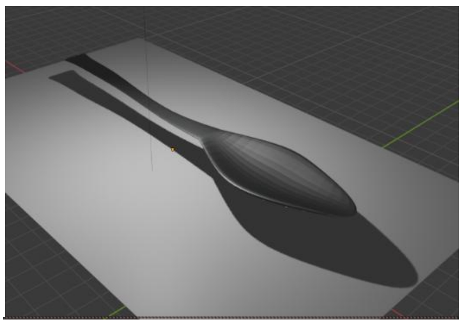
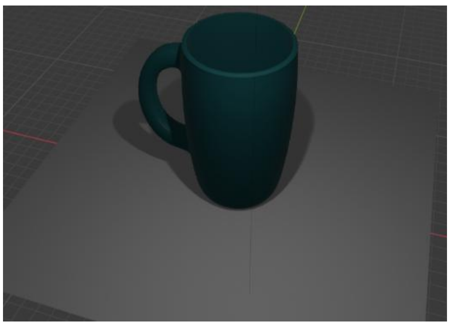
3. Work Done / Observations / Duties Performed

3.1 Work Done

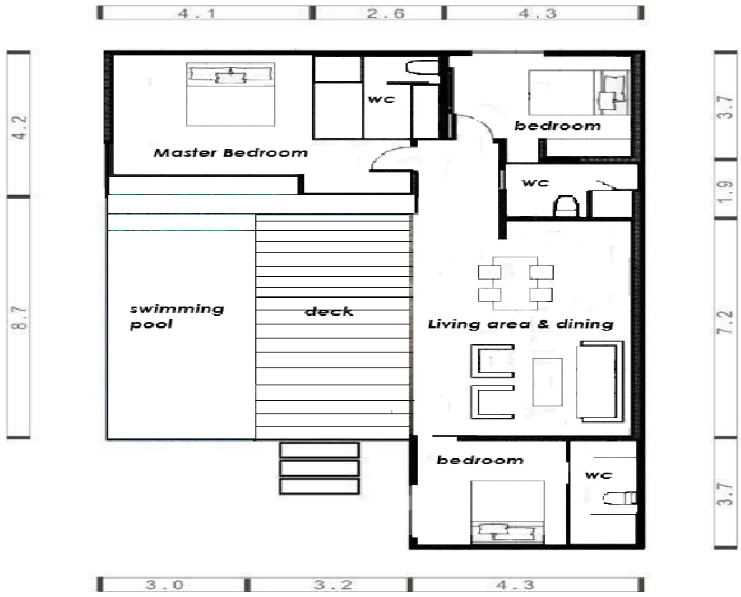
As part of my internship, I undertook an extensive 3D interior modelling project using **Blender**, a powerful open-source 3D creation suite. The project began with fundamental object modelling (Week 2) and culminated in a **detailed, room-by-room interior design of a complete house**, including architectural planning, material assignment, lighting, and spatial arrangement.

The work was divided into phases:

- **Basic Object Modelling (Week 2):**
    - Practiced core modelling techniques using real-world objects like a **hammer**, **headphones, pencil, sandals, mug , coin , low poly house**, and other small assets Focused on using **modifiers**, **shading**, and **bevelling**.

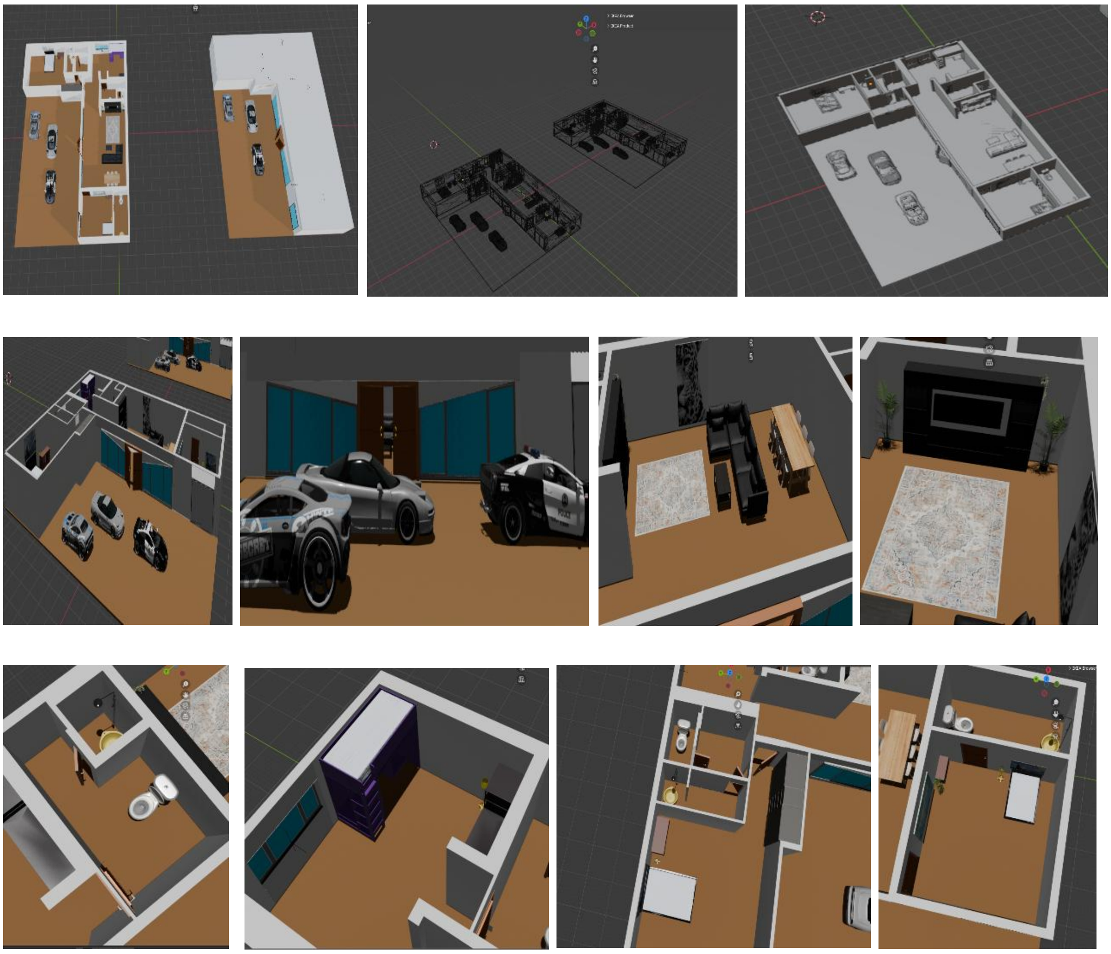
- **Floor Plan Design (Week 3 onwards):**
  - Created a custom **floor plan** for a modern residence from scratch.
  - Defined the spatial layout including **Master Bedroom, Kids Bedroom, Guest Bedroom, Hall/Living Room, and multiple Washrooms.**



- **Room-wise Modelling & Texturing (Main Project, week 4):**
  - Constructed each room with **realistic props**, ensuring distinct interiors:
    - **Master Bedroom:** Double bed with headboard, full-sized wardrobe, framed painting, lamps, and separate lighting.
    - **Guest Bedroom:** Neutral styling with simplified wardrobes, textures, and layout for minimalism.
    - **Living Hall:** L-shaped sofa, centre table, rug, **TV cabinet with emission shader**, and **showcase with decor items**.
    - **Washrooms: WC (commode)**, **partitioned shower area**,
  - Emphasized **material accuracy**: Used **glass shaders** for panel windows, **metal shaders** for handles, **wood and leather** for beds/wardrobes/sofas.

- Included multiple **window styles** in each room (sliders, panel windows, open hatches).
- Ensured proper **lighting setup**, reflections, and use of HDRI for environmental realism.

- **Scene Composition and Detailing:**
    - Modelled and placed **props** like wall paintings, showcase items, lights, rugs, and doors.
    - Used **Blender Add-ons** (free asset libraries like *BlenderKit* or *Sketchfab add-on*) to incorporate some base meshes and fine-tune them further.
    - Optimized render settings for fast yet detailed outputs.



3.2 Observations (Learning Curve Insights)

As a beginner stepping into Blender, I encountered common but important insights during the modelling journey:

- **Navigation Mastery:** The importance of understanding viewport controls, transform tools (G, S, R), and shortcut usage for efficient modelling.
- **Edit Mode vs Object Mode:** Understanding mesh-level editing versus object placement was crucial for avoiding workflow issues.
- **Snapping & Precision Modelling:** Initial models lacked alignment until snapping tools and grid overlays were used.
- **Modifiers as Game-Changers:** Discovery of **Mirror**, **Subdivision Surface**, and **Bevel modifiers** significantly reduced manual work.
- **Material Assignment Confusion:** Early confusion in texture mapping and assigning multiple materials to the same object led to unintentional overlapping until proper slot usage was learned.
- **Lighting Realism:** Realistic renders demand more than good models—HDRI, point lighting, and ambient light balance make a major difference.
- **Render Time vs Quality:** Balancing sample count, denoising, and resolution was a constant learning experience.
- **Asset Library Awareness:** Add-ons like *BlenderKit*, *Sketchfab Importer*, or *Archipack* added value by providing pre-modeled assets for inspiration or base editing.
- **Camera Work & Perspective:** Although cameras weren't widely used in this project, their impact on showcasing interior perspectives became evident.

3.3 Duties Performed

Throughout the internship, I actively engaged in the following responsibilities:

- Designed a complete **interior house model**, divided into individual, uniquely styled rooms.
- Drafted a **floor plan** for spatial arrangement, considering flow, realism, and accessibility.
- Modelled furniture and architectural elements from scratch (beds, wardrobes, sofas, windows, TV cabinets, carpets, washroom items).
- Applied **UV unwrapping** and texturing for realism in surface design.
- Managed **lighting and material** consistency throughout the project.
- Incorporated **add-on libraries** for asset acceleration, followed by customization.
- Captured and rendered multiple **perspective-based screenshots** for documentation and presentation.
- Used organizational principles to separate objects into collections, ensuring proper scene management and render control.
- Iteratively refined props and layout after feedback and self-evaluation.

4. Learning After Internship

This internship provided an enriching opportunity to explore the field of 3D design and animation using **Blender**, allowing me to apply digital modelling concepts to a realistic and creative interior design project. By working on the complete interior modelling of a residential house, I gained **hands-on experience with the entire 3D asset creation pipeline** — from base modelling and scene planning to texturing, lighting, and rendering. I significantly enhanced my understanding of **polygonal modelling**, **material assignment**, and **scene organization**. Learning how to work with modifiers like **Mirror**, **Subdivision Surface**, and **Bevel** helped me efficiently create detailed and symmetrical models. I also learned to use **PBR materials**, shaders, and glass textures to achieve realistic surfaces across different assets such as **beds, sofas, wardrobes, lamps, and bathroom fixtures**. One of the key skills I developed was the ability to **compose complex interior environments**, maintaining unique aesthetics for each room including **master bedroom**, **guest bedroom**, **kids' room**, **living hall**, and **washrooms**. I became confident in using Blender's **viewport navigation, snapping tools**, and **collection system** for organized workflow management. Furthermore, I gained practical experience with **render settings**, camera perspectives, lighting setups, and **HDRI environments** — all of which are crucial in creating a photo-realistic final output. Understanding **UV unwrapping and texture mapping** allowed me to control how materials appear on different surfaces. Another important aspect of this internship was the opportunity to apply **creative problem-solving** while optimizing models, balancing detail with performance, and making design decisions that reflect real-life interior planning. Working with limited camera setups encouraged me to explore composition techniques and better frame my scenes. In addition, I learned the value of **iterative refinement**, where continuous observation and minor adjustments lead to a polished result. Creating props like **TV cabinets, wall paintings, rugs, and even sanitary items** gave me a deep appreciation of detail-oriented modelling.

Overall, this experience strengthened both my **technical proficiency** and **visual design sensibility**, and it has inspired me to pursue further learning in **3D design, architectural visualization**, and potentially, **game asset creation**. Blender has gone from being a complex tool to a creative outlet where I can bring ideas to life.

5. Summary / Conclusion

This internship with Infosys gave me hands-on experience in 3D modelling using Blender. Over the course of the internship, I started from the basics and progressed to designing a complete house interior model. I learned how to use essential tools like modifiers, shaders, lighting, materials, and camera views. I also explored free Blender add-ons for assets such as furniture, decor items, and lighting elements.

Through this journey, I created detailed models including doors, beds, wardrobes, paintings, carpets, TV cabinets, sofas, lamps, and bathroom elements like WC, showers, and buckets. Each room—master bedroom, guest bedroom, kids' room, hall, and washrooms—was designed with distinct themes, materials, and layout. I also built the base floor plan myself and used various window types with glass and panel materials.

This internship helped me understand the workflow of modelling, the importance of scene organization, and how to bring realistic details into a project. I faced and overcame beginner challenges like mesh clipping, incorrect lighting, and object alignment. Overall, it boosted my confidence in using Blender for professional 3D modelling and gave me practical experience in turning ideas into detailed digital environments.

6. Future Scope

This 3D modelling internship has laid a strong foundation in Blender, and there is significant scope for further growth and improvement. The project can be enhanced in the following ways:

1. **Animation & Walkthroughs**: Creating animations or virtual walkthroughs of the house interior can add an interactive element for presentations or architectural use.
2. **Game Engine Integration**: The models can be imported into game engines like Unity or Unreal Engine for interactive 3D experiences or VR applications.
3. **Asset Optimization**: Optimizing mesh topology and using low-poly techniques can help make the models game- or web-ready.
4. **Real-World Architectural Standards**: Applying proper architectural measurements and furniture layouts to match real-world dimensions can help make the models useful for interior design clients.

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3. **CG Geek** (YouTube Channel) – https://www.youtube.com/user/Blenderfan93
   (Tutorials and creative project inspirations)
4. **Sketchfab** – https://sketchfab.com
   (Reference models for understanding detailing and realistic materials)
5. **Poly Haven** – https://polyhaven.com
   (Free HDRIs, textures, and 3D assets used for realistic environments)
6. **Blender Artists Forum** – https://blenderartists.org
   (Community discussions and problem-solving related to Blender use)
7. **Blender Market** (Free Add-ons) – https://blendermarket.com
   (Free or open-source add-ons used for furniture and asset modelling)